Math 110 — Week 5

Solutions

HW5 P1.

Idea. The null space is given by two zeros, so factor. Surjectivity: find polynomials sent to e_1 and e_2. Those same polynomials give the basis of the quotient.

1. (a) $p \in \operatorname{null} T$ means $p(0) = p(1) = 0$, i.e. $x(x-1)$ divides p. With $\deg p \le 3$: $p = x(x-1)(ax+b)$. So $\operatorname{null} T = \operatorname{span}\big(x(x-1),\ x^2(x-1)\big)$, of dimension 2.
2. $T(1-x) = (1,0)$ and $T(x) = (0,1)$. These span $\mathbb{R}^2$, so T is surjective. (Check: $4 = 2+2$.)
3. (b) $\widetilde{T}\colon \mathcal{P}_3(\mathbb{R})/\operatorname{null} T \to \mathbb{R}^2$, $\widetilde{T}(p + \operatorname{null} T) = (p(0), p(1))$. It is well-defined because two names of the same set differ by a polynomial vanishing at 0 and 1.
4. (c) $\widetilde{T}$ is an isomorphism, and $\widetilde{T}\big((1-x) + \operatorname{null} T\big) = e_1$, $\widetilde{T}\big(x + \operatorname{null} T\big) = e_2$. The inverse of an isomorphism sends a basis to a basis, so $(1-x) + \operatorname{null} T,\ x + \operatorname{null} T$ is a basis of the quotient.

HW5 P2.

Idea. The formula is forced, so uniqueness is immediate. Existence is the well-definedness check, and it uses exactly $U \subseteq \operatorname{null} T$.

1. *Uniqueness.* Every element of V/U is $v + U$ for some v, and the condition prescribes the value Tv there. So at most one such map exists.
2. *Well-defined.* If $v + U = w + U$, then $v - w \in U \subseteq \operatorname{null} T$, so $Tv - Tw = T(v-w) = 0$.
3. *Linear.*
$$\widetilde{T}(\lambda(v+U) + (w+U)) = \widetilde{T}((\lambda v + w) + U) = T(\lambda v + w) = \lambda \widetilde{T}(v+U) + \widetilde{T}(w+U).$$
4. *Null space.* $\widetilde{T}(v+U) = 0 \iff Tv = 0 \iff v \in \operatorname{null} T$. So $\operatorname{null} \widetilde{T} = \{v + U : v \in \operatorname{null} T\} = (\operatorname{null} T)/U$.

HW5 P3.

Idea. If φ takes one nonzero value, scaling the input produces every value.

1. Suppose $\varphi \neq 0$. Choose v with $\varphi(v) = c \neq 0$.
2. For any $a \in \mathbb{F}$:
$$\varphi\big(\frac{a}{c} v\big) = \frac{a}{c}\varphi(v) = a.$$
So φ is surjective.

HW5 P4.

Idea. φ_j reads off the j-th coordinate. Functionals are compared by evaluating on the basis.

1. (a) Write $v = \sum_k a_k v_k$. Then $\varphi_j(v) = \sum_k a_k \varphi_j(v_k) = a_j$. Substituting $a_j = \varphi_j(v)$ gives the formula.
2. (b) Evaluate the right side at v_k: $\sum_j \psi(v_j)\varphi_j(v_k) = \psi(v_k)$. So the two functionals ψ and $\sum_j \psi(v_j)\varphi_j$ agree on a basis of V, hence are equal.

HW5 P5.

Idea. Column j of $\mathcal{M}(T')$ is the coordinate vector of $T'(w_j')$, and coordinates of a functional are its values on $v_1, \dots, v_n$.

1. Let $B = (v_1, \dots, v_n)$, $C = (w_1, \dots, w_m)$, $A = (a_{lk})$, so $Tv_k = \sum_l a_{lk} w_l$.
2. Evaluate $T'(w_j')$ on the basis vector v_k, using column k of $\mathcal{M}(T)$:
$$T'(w_j')(v_k) = w_j'(Tv_k) = w_j'\Big(\sum_l a_{lk} w_l\Big) = a_{jk}.$$
3. By P4(b),
$$T'(w_j') = \sum_k T'(w_j')(v_k)\, v_k' = \sum_k a_{jk} v_k'.$$
4. So the (k, j) entry of $\mathcal{M}(T')$ is a_{jk}: $\mathcal{M}(T') = A^{\mathsf{T}}$.

HW5 P6.

Idea. A row is in U^0 when it gives 0 on each of the two spanning vectors.

1. $(a\ b\ c)$ applied to $(1, 1, 0)$ is $a + b$; applied to $(0, 1, 1)$ is $b + c$. Both zero: $a = -b$, $c = -b$.
2. $U^0 = \operatorname{span}\big((1\ \ -1\ \ 1)\big)$, the functional $(x, y, z) \mapsto x - y + z$.
3. $\dim U = 2$ (the two vectors are not multiples of each other), $\dim U^0 = 1$, and $2 + 1 = 3$.

HW5 P7.

Idea. Injective: for $v \neq 0$ build a functional with $\varphi(v) = 1$ by making v a basis vector.

1. j_V is linear:
$$j_V(\lambda v + w)(\varphi) = \varphi(\lambda v + w) = \lambda\varphi(v) + \varphi(w).$$
2. Let $v \neq 0$. Extend v to a basis of V and let φ be 1 on v and 0 on the other basis vectors. Then $j_V(v)(\varphi) = 1$, so $j_V(v) \neq 0$. Hence $\operatorname{null} j_V = \{0\}$.
3. If $\dim V = n < \infty$: $\dim V'' = \dim V' = n$. An injective map between spaces of the same finite dimension is an isomorphism.

R1.

Idea. Same map as HW5 P1.

1. (a)
$$E(\lambda p + q) = (\lambda p(0) + q(0), \lambda p(1) + q(1)) = \lambda E(p) + E(q).$$
$E(1 - x) = (1, 0)$, $E(x) = (0, 1)$: surjective.
2. (b) $U = \{x(x-1)(ax+b)\}$, basis $x(x-1),\ x^2(x-1)$ (different degrees, and $\dim U = 4 - 2 = 2$).
3. (c) Well-defined: if $p + U = q + U$ then $p - q \in U = \operatorname{null} E$, so $E(p) = E(q)$. Linear because E is. Injective:
$$\widetilde{E}(p + U) = 0 \Rightarrow p \in U \Rightarrow p + U = 0 + U.$$
Surjective because E is.
4. (d) $(1 - x) + U \mapsto (1, 0)$ and $x + U \mapsto (0, 1)$. These two elements form a basis of V/U because their images form a basis and $\widetilde{E}$ is an isomorphism.

R2.

Idea. Everything follows from $(T^*\varphi)(v) = \varphi(Tv)$.

1. (a) $T^*(\varphi) = \varphi \circ T$ for $\varphi \in W^*$. Linear:
$$T^*(\lambda\varphi + \psi) = (\lambda\varphi + \psi) \circ T = \lambda(\varphi \circ T) + \psi \circ T.$$
2. (b) For $\varphi \in Z^*$:
$$(S \circ T)^*\varphi = \varphi \circ S \circ T = (S^*\varphi) \circ T = T^*(S^*\varphi).$$
3. (c) Suppose $T^*\varphi = 0$, i.e. $\varphi(Tv) = 0$ for all v. Since T is surjective, every $w \in W$ is some Tv, so $\varphi(w) = 0$ for all w. So $\varphi = 0$, and T^* is injective.
4. (d) $(T^*\ell)(x, y) = \ell(2x - y) = 6x - 3y$.

E1.

Idea. Rows of the inverse of the matrix whose columns are the basis.

1. $M = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $\det M = -2$,
$$M^{-1} = \frac{1}{-2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}.$$
2. $\varphi_1 = (\frac{1}{2} \;\; \frac{1}{2})$, $\varphi_2 = (\frac{1}{2} \;\; -\frac{1}{2})$. Check: $\varphi_1(v_1) = 1$, $\varphi_1(v_2) = 0$, $\varphi_2(v_1) = 0$, $\varphi_2(v_2) = 1$.
3. The coordinates of ψ in the dual basis are its values on the basis:
$$\psi = \psi(v_1)\varphi_1 + \psi(v_2)\varphi_2 = 4\varphi_1 + 2\varphi_2.$$
Check: $4(\frac{1}{2} \;\; \frac{1}{2}) + 2(\frac{1}{2} \;\; -\frac{1}{2}) = (3 \;\; 1)$.